

Max time: 10 Min

ID No:

Name:

Max Marks: 08

Sec No.:

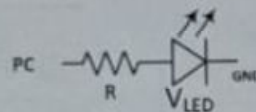
Note: No credit for any answer without proper justification

1. From which address(s) is the CS of the NMI handler fetched by the 8086 processor? (2M)

1M 0000AH
1M 0000BH

given if 000A^h is written or even if 000B
(1M) 000A/000A is written

2. The base address of 8255 is B7h (Use B7h and next possible addresses in increasing sequence for 8255). You are required to switch on an LED (as shown below) directly connected to PC2 of 8255 interfaced to 8086 using BSR mode.



- a. Fill in the blanks with 2 lines (instructions) of code (only two instructions- no data declarations allowed) to configure 8255 in BSR mode

mov al, 10000000b [1M] (*take all don't care's as 0, mention values in hex only)

out FFh, al [1M]

- b. Fill in the blanks with 2 lines of code that can be used to switch on LED connected to port PC2 (only two instructions- no data declarations allowed)

mov al, 00000101b [1M] (*take all don't care's as 0, mention values in hex only)

out FFh, al [1M]

- c. What is the address of Port B *BFh* [1M]

- d. Which address line from 8086 will be connected to the following A1 pin of 8255. *A6* [1M]

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
1 - I/O Mode	Port A Mode		Port A	Port C	Port B	Port B	Port C
	00 - Mode 0		1 - I/p	Upper	Mode	1 - I/p	Lower
	01 - Mode 1		0 - o/p	1 - I/p	0 - Mode 0	0 - o/p	1 - I/p
	1x - Mode 2			0 - o/p	1 - Mode 1		0 - o/p
Group A				Group B			

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀						
0 - BSR	x	x	x	Bit ₂	Bit ₁	Bit ₀	Bit Set/Reset						
Don't Care Condition				PC	0	1	2	3	4	5	6	7	1 - Set 0 - Reset
				B ₀	0	1	0	1	0	1	0	1	
				B ₁	0	0	1	1	0	0	1	1	
				B ₂	0	0	0	0	1	1	1	1	

1011 0111 - PA
1011 1111 - PB